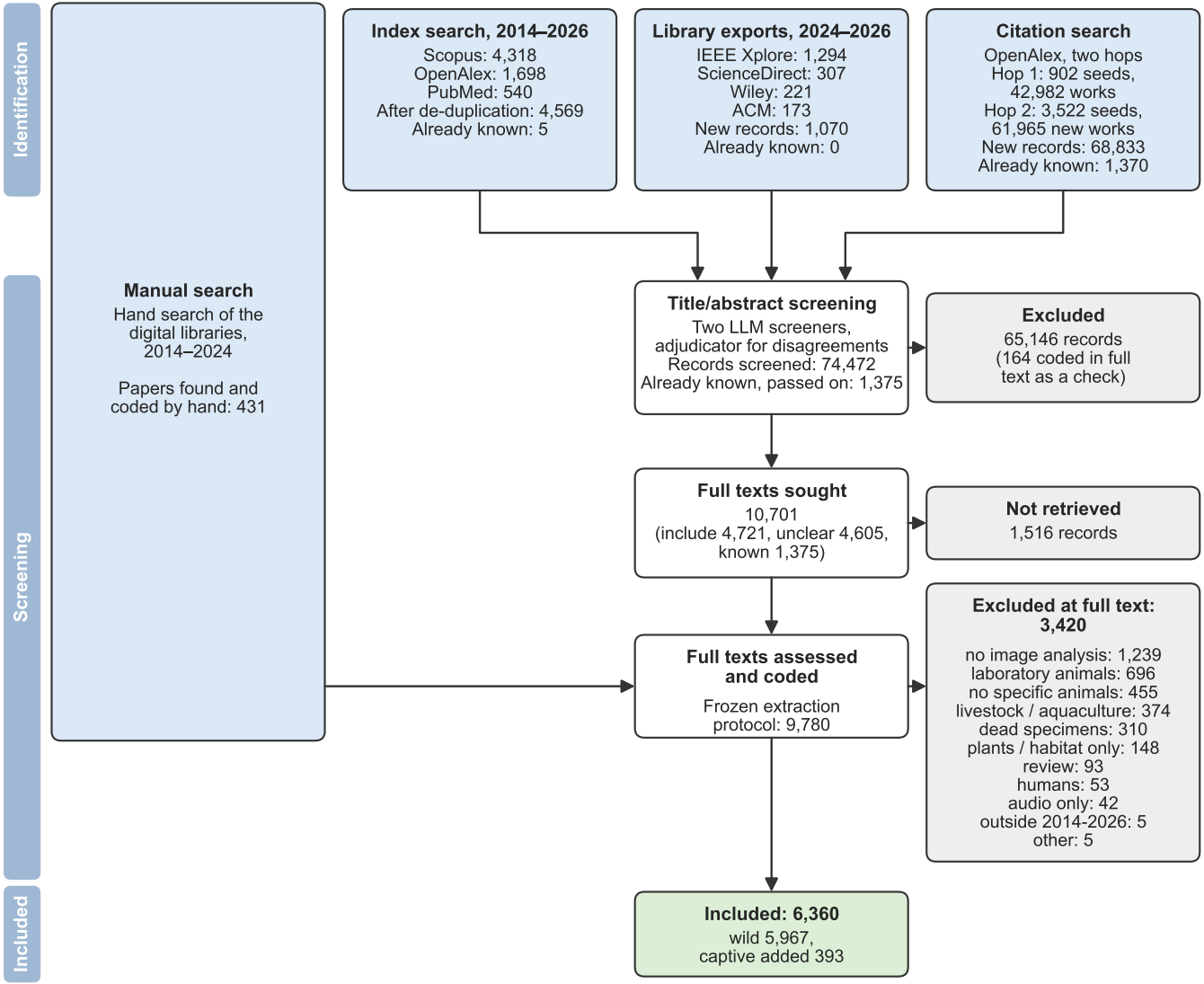
Identification
Screening
Included
Manual search
Hand search of the digital libraries, 2014–2024
Papers found and coded by hand: 431
Index search, 2014–2026
Scopus: 4,318
OpenAlex: 1,698
PubMed: 540
After de-duplication: 4,569
Already known: 5
Library exports, 2024–2026
IEEE Xplore: 1,294
ScienceDirect: 307
Wiley: 221
ACM: 173
New records: 1,070
Already known: 0
Citation search
OpenAlex, two hops
Hop 1: 902 seeds, 42,982 works
Hop 2: 3,522 seeds, 61,965 new works
New records: 68,833
Already known: 1,370
Title/abstract screening
Two LLM screeners, adjudicator for disagreements
Records screened: 74,472
Already known, passed on: 1,375
Excluded
65,146 records
(164 coded in full text as a check)
Full texts sought
10,701
(include 4,721, unclear 4,605, known 1,375)
Not retrieved
1,516 records
Full texts assessed and coded
Frozen extraction protocol: 9,780
Excluded at full text: 3,420
no image analysis: 1,239
laboratory animals: 696
no specific animals: 455
livestock / aquaculture: 374
dead specimens: 310
plants / habitat only: 148
review: 93
humans: 53
audio only: 42
outside 2014-2026: 5
other: 5
Included: 6,360
wild 5,967,
captive added 393